When Certification Hurts: Accreditation Signals and Graduate Wages

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April 25, 2026

[Preliminary draft, do not cite]

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Employers cannot directly observe program quality, and institutional reputation provides reliable signals for only a small fraction of graduates. We study whether program-level accreditation in Chile affects graduate wages, using a staggered difference-in-differences design applied to administrative earnings records for approximately 2,000 programs and exploiting first-time accreditation decisions between 2007 and 2018. After accreditation, the ability composition of graduating cohorts shifts upward while the formal employment rate falls. The average wage effect is approximately 3 percent, concentrated at longer horizons when higher-ability cohorts begin graduating. The wage response varies systematically with institutional reputation and accreditation intensity, increasing by 9.1 percent at lower-reputation institutions and decreasing by 4.8 percent at elite institutions. The wage response is largest where prior information about the program was scarce, and the accreditation term was short.

Keywords: Program accreditation, employer learning, signaling, wage returns, quality disclosure

JEL: I23, I28, D83, J24, J31

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1 Introduction

Consider the information problem facing a recruiter who receives a resume from a graduate of a program they have never encountered. A business degree from a regional university, perhaps, or an engineering credential from one of the hundreds of private institutions that have entered the market in recent decades. The recruiter cannot inspect the curriculum, cannot verify faculty quality, and cannot observe the underlying productivity of the candidate. Institutional reputation fills this gap, but only partially since it functions as a reliable signal for a handful of long-standing elite universities and provides almost nothing for the rest of the distribution (MacLeod and Urquiola, 2015). For the vast majority of graduates, employers and workers are left in a state of mutual informational deficit. Third-party certification is the institution designed to resolve it. Whether real-world accreditation systems actually do so, however, is far from obvious.

If accreditation provides new information to employers, its effect on graduate wages should depend on how much employers already knew. When employers aggregate noisy proxies for worker productivity, such as institutional prestige, prior hiring experience, and alumni networks, their confidence in those estimates varies systematically across the market. At programs where these proxies already yield high-confidence predictions, a new public certification adds little marginal information. At programs where priors are weak or dispersed, a verifiable quality label resolves genuine uncertainty and may allow the market to update its valuation of the graduate (Altonji and Pierret, 2001). The implication is that accreditation's effect is not uniform. It should depend on the precision of what employers already believed. Whether this is what happens in actual labor markets, and with what magnitude, requires direct empirical evidence.

We investigate four questions. First, does first-time program accreditation change the size and composition of graduating cohorts? If accreditation attracts different students into the program, any subsequent change in average wages may reflect who graduates rather than how the labor market values them. Second, what is the effect of first-time program accreditation on the average wages of formally employed graduates? This establishes the aggregate wage response to the signal. Third, does the wage response vary with institutional reputation, which proxies for how much employers already knew about the program? Fourth, does it vary with the accreditation term awarded to the program, which captures what the accreditation signal conveyed? Together, the third and fourth questions test whether the wage response depends on the informational context in which the accreditation decision is received.

We answer these questions using Chile's higher education system between 2007 and 2018, which provides a well-suited setting for this analysis. There is high institutional heterogeneity, a centralized and transparent accreditation registry, and rich administrative data linking graduation records from the Ministry of Education to private-sector earnings reported to the tax authority. This yields a panel of nearly 18,000 program-cohort observations across approximately 2,000 degree programs. The identification strategy exploits the staggered adoption of first-time accreditation decisions. Programs choose when to seek evaluation, but the final decision (both the binary outcome and the duration awarded) is made by an independent panel of commissioners and is largely unanticipated by employers at the time of hiring (Barroilhet et al., 2022). We apply the imputation estimator of Borusyak et al. (2024) to address the negative-weight problems that arise in staggered designs under treatment effect heterogeneity. Pre-trend tests confirm that wages at treated and control programs followed identical trajectories before accreditation ($p > 0.60$ across all three wage horizons).

The graduate pool changes after accreditation. Total graduation counts rise, and the ability composition of graduates shifts upward as the share of high-ability entrants increases significantly, while the socioeconomic composition remains unchanged. These compositional changes begin appearing in graduating cohorts several years after the accreditation decision, consistent with the time required for newly enrolled students to complete their degrees.

The employment rate of the graduate pool falls following accreditation. This decline does not indicate that accreditation worsens labor market outcomes. The number of employed graduates rises, but less rapidly than the total number of graduates, pulling the employment rate down. The additional completers who graduate after accreditation may also be more likely to enter self-employment, the informal sector, or public-sector positions that fall outside the private-sector payroll records, so the formal-employment count rises at a slower pace than total graduates.

The average wage effect across all post-accreditation cohorts is approximately 3 percent, reaching statistical significance in years two and three after graduation. The dynamic profile reveals that this average masks variation over time. In the early post-accreditation window, when all graduates enrolled before accreditation was announced and the composition of the graduating cohort has not yet changed, the wage estimates are close to zero. At longer horizons, where higher-ability cohorts begin graduating, the estimates rise to 6 to 15 percent. The aggregate data cannot determine how much of the late-window effect reflects employer responses to the accreditation signal versus the mechanical consequence of higher-ability graduates earning more.

The heterogeneity results reveal systematic variation along two dimensions. When stratified by institutional reputation, graduates of programs at *Baseline* institutions experience wage increases of 9.1 percent in year one, while graduates of programs at *Top-tier* institutions experience wage reductions of 4.8 percent. When stratified by program accreditation intensity, programs receiving a short first-accreditation term show wage increases of 12.3 percent, while programs receiving a long term show no significant change. The wage response is largest where prior information about the program was scarce and the accreditation term was short, and it is negative or absent where prior information was abundant.

These findings make three contributions. First, we provide the first estimates of the effect of program-level accreditation on graduate wages in the labor market. The closest prior work, González-Velosa et al. (2015), estimates a 9.2 percentage-point wage differential per additional year of institutional accreditation using cross-sectional OLS, a design that cannot separate the effect of the label from pre-existing quality differences between programs. By exploiting variation in the timing of first-time accreditation decisions, we document both the average wage effect and the compositional changes that accompany it (Dale and Krueger, 2002; Hastings et al., 2013; Hoekstra, 2009; MacLeod et al., 2017; Reyes et al., 2016). Beyond the average effect, we show that the ability composition of graduating cohorts shifts upward following accreditation while the socioeconomic composition remains stable, and the formal employment rate falls as the expanded graduation pool includes marginal completers.

Second, we document that the wage response varies systematically with institutional reputation and program accreditation intensity. The pattern is consistent with the wage effect depending on how much new information the accreditation decision provided relative to what employers already knew. This connects the accreditation setting to the broader literature on reputation and third-party certification (Dranove and Jin, 2010; Dranove et al., 2003; Elfenbein et al., 2015; Jin, 2005; Lizzeri, 1999; MacLeod et al., 2017; MacLeod and Urquiola, 2015), and to the empirical literature on employer learning and the role of public signals in labor markets (Altonji and Pierret, 2001; Farber and Gibbons, 1996; Lange, 2007; Spence, 1973).

Third, the heterogeneity results carry distributional implications. Because the wage response is concentrated among graduates of programs at lower-reputation institutions, the benefits of accreditation flow disproportionately to graduates whose credentials would otherwise be discounted by employers. In higher education systems characterized by significant institutional heterogeneity, a common feature of education markets across the developing world, formal certification may function as a mechanism for reducing the informational disadvantage that graduates at the bottom of the reputation distribution face. The aggregate design of this paper cannot establish the individual-level channels through which this occurs, and the extensions required to do so are discussed in Section 7.

The remainder of the paper proceeds as follows. Section 2 describes the institutional background. Section 3 presents the data sources and sample construction. Section 4 describes the identification strategy and the limitations of the research design. Section 5 presents the graduation count, composition, and wage results. Section 6 examines how the wage effects vary with institutional reputation and program accreditation intensity. Section 7 concludes.

2 Institutional Background

2.1 The Accreditation System

Chile's higher education system encompasses hundreds of degree programs offered by institutions ranging from long-established public universities with decades of employer relationships to private providers with little observable track record. For graduates of elite programs, the institutional brand functions as a reliable proxy. Employers have decades of hiring experience and can form confident predictions about productivity. For graduates of the many programs where that reputational history is thin or absent, no equivalent shortcut exists. The graduate knows what their training was worth, but the employer does not. Without a way to distinguish quality, the market treats graduates of unrecognized programs as interchangeable and wages compress toward the average (Akerlof, 1970; MacLeod et al., 2017). Programs that invest in quality cannot capture the return because employers cannot observe the investment.

Chile's 2006 Quality Assurance Act created the National Accreditation Commission (CNA) to address this problem. CNA conducts two separate evaluations. Institutional accreditation evaluates the university as an entity and is strongly correlated with measures of selectivity and prestige, making it a natural proxy for the precision of employer prior beliefs about programs at that institution (Barroilhet, 2019; Bordon and Fu, 2015). Program-level accreditation evaluates each degree program individually. A university can have an accredited law program and an unaccredited business program in the same building. The program-level accreditation decision therefore conveys two layers of information simultaneously. The first is a binary quality floor, indicating that the program has demonstrated compliance with at least the minimum standards set by the Commission. The second is the term duration, which locates the program on a quality gradient above that floor. A short term signals marginal compliance, and a longer term signals stronger performance across the Commission's evaluation criteria. Both layers arrive at the same moment, but they carry different informational content. For programs at institutions where employers held diffuse priors, the binary floor alone resolves substantial uncertainty. For programs at well-known institutions, the binary floor is expected and uninformative; only the term duration, and whether it meets or falls short of what the institutional context would suggest, carries new information for the employer. Accreditation is mandatory only for programs in medicine, dentistry, and education, where it is tied to professional licensing. We exclude these three fields from the analysis. For all remaining fields, accreditation is a voluntary strategic decision.

These two evaluations do not perfectly correlate. Table 2 reports the joint distribution of treated programs by institutional accreditation tier and program first-accreditation term. While programs at Baseline institutions are concentrated in the Low-term category (58.3 percent receive one to three years), nearly 30 percent of programs at Top-tier institutions also received a Low first-accreditation term. A program at a prestigious institution can receive a term that falls short of what the institution's reputation might suggest. The heterogeneity analysis in Section 6 exploits variation along both dimensions, where institutional tier captures the employer's prior and program term duration captures the content of the new signal.

2.2 Accreditation as an Information Shock

When CNA grants accreditation, the decision is published immediately in a public registry integrated into the government's official higher education information portal.[1] Employers screening candidates can consult it directly. The ruling, therefore, changes what a recruiter can verify about a program the morning after the Commission's decision. The timing of this change matters for wage outcomes. In Chile, graduation ceremonies occur at the end of the academic year. A program accredited mid-year will have a fraction of that year's graduates already entering the labor market carrying the new label, but the first cohort for whom the accreditation signal is observable throughout their job search is the one graduating in the year following the award.

A key feature of this setting is that employers are not tracking the accreditation pipeline. Programs submit applications and undergo evaluation over several months, but there is no public disclosure of a program's application status or the contents of the peer review process. Employers have no reason to anticipate which programs are under evaluation or when a ruling will arrive. The registry entry, therefore, functions as a genuine information shock from the employer's perspective, even though the program's decision to apply was voluntary and strategic. Furthermore, the presence of accreditation status in official government portals does not guarantee that employers actively search for it. Whether employers update wages in response to the signal at all is not obvious in advance. It is an empirical question our analysis answers.

Because programs decide when to seek their first accreditation based on their own assessment of readiness and competitive pressure, first-time accreditation decisions are distributed across the 2007 to 2018 sample period in a pattern driven by program-level strategy rather than any government-mandated schedule. This staggered voluntary adoption generates variation in the timing of the information shock that, from the employer's perspective, is effectively unpredictable. The endogeneity concern is about timing, as programs apply when they believe they are likely to pass, so treated programs may differ from never-treated ones in ways correlated with wages. This concern motivates robustness specifications that separately employ not-yet-accredited and never-accredited programs as comparison groups, each isolating a different dimension of the selection problem. What is not endoge-

[1] Accreditation status by program is publicly searchable at https://www.mifuturo.cl/buscador-de-acreditacion/ (SIES, Ministerio de Educación de Chile).

nous is the ruling itself. Once an application is filed, an independent panel of CNA commissioners issues a decision on both the binary outcome and the term length, and programs cannot control what that decision will be. The identification strategy, developed in Section 4, exploits the staggered timing of first accreditation as a source of variation that is unpredictable from the employer's side, relying on parallel trends across programs with different accreditation timing.

3 Data

3.1 Labor Market Outcomes

Formal private-sector employers in Chile report annual earnings to the Internal Revenue Service (SII) for every worker on payroll. Under a data access agreement with the Ministry of Finance (*Ministerio de Hacienda*), we obtained access to SII earnings records linked to higher education graduation files. Prior work using this data to study returns to higher education in Chile includes Hastings et al. (2013), González-Velosa et al. (2015), and Bordon and Fu (2015). Graduates employed in the public sector, the self-employed, and the informal sector fall outside the SII universe and are not observed. Public-sector employment and wage records are maintained separately by the Budget Office (DIPRES) and the Comptroller General, and are not linked to the SII administrative data used in this study. If public-sector employers condition hiring on accreditation status, accreditation may shift graduates across sectors in ways that the private-sector wage sample cannot capture. The direction of the resulting bias depends on which graduates move. If accreditation enables high-ability graduates to enter public-sector positions they would not otherwise access, the private-sector sample retains a less favorably selected subset, and the estimated wage effect understates the full labor market return. Extending the analysis to include public-sector wages would require a separate inter-institutional data agreement.

For each program–graduation year cell, we observe the number of graduates matched to formal employment and their mean annual taxable earnings one, two, and three years after graduation. Because the data are aggregated to program-cohort means, the analysis cannot examine how accreditation affects the within-cohort distribution of wages. Whether accreditation compresses wage dispersion among graduates of the same program or shifts the entire distribution uniformly is a question that requires individual-level wage records and is left for future work. The three-horizon structure is a design feature. Measuring wages at multiple post-graduation horizons allows us to examine whether the wage effect changes as employers accumulate direct experience with graduates from newly accredited programs.

3.2 Accreditation Timing

The CNA maintains a public registry that records, for each program, the exact date of the Commission's ruling, the binary accreditation outcome, and the duration of the term awarded. For each accredited program, we record the year of first CNA accreditation as the treatment date G_p and construct event time $\kappa_{pt} = t - G_p$ to position each graduation cohort relative to the moment the signal became publicly verifiable. First-time accreditations are distributed across all years from 2007 to 2018, providing the cross-cohort staggering that the imputation estimator exploits.

The registry also records the length of each accreditation term, ranging from two to seven years. We use modal institutional term length to construct the three-tier quality proxy (Baseline: two to three years; Enhanced: four to five years; Top-tier: six to seven years) that drives the heterogeneity analysis in Section 5. Programs' first-time term length serves a parallel role at the program level, measuring the intensity of the quality signal the Commission sent at the moment of first accreditation and allowing us to test whether programs that received longer initial terms generate correspondingly larger wage effects for their graduates.

3.3 Student Composition

Any effects on wages estimated in this paper could, in principle, reflect two distinct mechanisms. Employers may be updating their beliefs about program quality in response to the accreditation signal, or high-ability students may be selecting into newly accredited programs and mechanically raising average wages through their own human capital. Separating these channels requires data on who attends each program in each cohort. That information comes from DEMRE, Chile's centralized admissions authority, which assigns students to programs through a deferred acceptance algorithm based on standardized test scores (*Prueba de Selección Universitaria*, PSU) and secondary school GPA.

When students register to sit the PSU, they report family income, parental education, and prior academic performance. Because registration is universal among applicants to the centralized system, these records are available at the individual level and can be aggregated to the program–admission cohort level, where they link to graduation cohorts. The composition test in Section 5.2 uses this link to ask whether accreditation shifts the socioeconomic profile of students graduating from treated programs. If enrollment responds to accreditation at the time of the decision, graduation composition should begin shifting approximately one degree-length later, consistent with the enrollment responses documented directly by Molina and Valdebenito (2026). Because socioeconomic characteristics are measured at registration rather than at graduation, the test assumes that pre-enrollment SES remains informative about students' backgrounds by the time they enter the labor market, which is a reasonable assumption given that parental education and family income at age 18 are largely determined before university begins.

3.4 Sample Construction

The full wage panel contains 25,054 program–graduation year observations from 3,429 programs. Two restrictions define the eligible sample. First, we exclude medicine, dentistry, and education, where accreditation is mandatory and tied to professional licensing rather than voluntary quality signaling, as discussed in Section 2. Second, we restrict to programs offered by institutions that held institutional accreditation at some point during the sample period. Employers screening graduates from unaccredited institutions face a different inference problem: the absence of an institutional track record confounds the program-level signal, and students at unaccredited institutions are ineligible for public higher education funding, introducing a selection channel qualitatively distinct from the sorting observed within the accredited sector.

The treatment window covers first accreditations between 2007 and 2018. The lower bound reflects data availability, as 2007 is the first year for which graduation cohorts can be reliably linked to wage outcomes. The upper bound reflects an institutional discontinuity: undergraduate program-level accreditation was suspended in 2018, making it the last year in which a first accreditation can be interpreted as a clean voluntary information shock. Programs with no accreditation event within this window form the never-treated control group. These restrictions yield an estimation sample of nearly 18,000 observations across roughly 2,000 programs.

3.5 Descriptive Statistics

Table 1 compares pre-accreditation cohort characteristics across treated and control programs, restricting to observations before each program's first accreditation event ($\kappa < 0$). The comparison covers the 1,154 treated programs with at least one pre-accreditation wage observation. All 773 never-treated programs are included.

Panel A reveals that graduates of control programs earn approximately 25 percent more per year than graduates of treated programs (CLP 7.5 million versus CLP 6.0 million), while being matched to formal employment at slightly lower rates (79.8 percent versus 82.1 percent). The wage gap does not reflect a student quality advantage in the control group. Panel B shows that treated programs enter the sample with modestly stronger student bodies. Their graduates score higher on the PSU and on secondary school GPA, and come from slightly higher-SES households, with a larger share of college-educated parents (45.1 percent versus 40.2 percent). The wage gap instead reflects institutional composition. Panel C shows that control programs are disproportionately drawn from Top-tier institutions (16.0 percent versus 10.3 percent), programs at well-established universities whose graduates carry a recognizable brand that substitutes for accreditation in employer priors. Treated programs concentrate in Enhanced institutions (51.6 percent), universities with solid but nationally unrecognized reputations, for whom the CNA stamp closes a genuine information gap. They are also larger (32 versus 24 graduates per cohort) and graduate a substantially higher share of women (59 percent versus 46 percent), reflecting which sectors pursued voluntary accreditation early.

These baseline differences are not a threat to identification. They motivate it. The event-study design absorbs all time-invariant program differences through unit fixed effects and identifies the accreditation effect from within-program wage changes around the treatment event. The descriptive table shows that cross-sectional comparisons between treated and control programs would be confounded by the very reputational differences that the analysis places at its center. The within-program design sidesteps that confound, and the composition data in Panel B provide the baseline against which post-accreditation sorting can be assessed.

4 Empirical Strategy

4.1 Identification and Research Design

The central identification challenge is not establishing that accreditation happened but establishing that its timing was unanticipated. If programs knew years in advance exactly when they would receive their first accreditation, they could invest strategically during the pre-accreditation period, and any post-accreditation wage gains would reinforce the quality signal from earlier quality investments. The staggered difference-in-differences design we use below is built precisely to handle this timing uncertainty.

We estimate the effect of first-time program accreditation on graduate wages using a staggered difference-in-differences design. The treatment is defined as a binary indicator for the year a program receives its initial accreditation from the National Accreditation Commission (CNA). Our primary outcomes are the log annual earnings of graduates measured one, two, and three years after labor market entry.

The relevant treatment timing depends on the graduation cohort relative to the public announcement. We define event time κ_{pt} as the difference between graduation year t and the year of first accreditation G_p:

$$\kappa_{pt} = t - G_p \tag{1}$$

where $G_p = \infty$ for never-treated programs. This specification captures the information available to employers at the time of hiring. Because firms observe the accreditation status of the degree-granting program, the wage effects should accrue to cohorts graduating on or after the date the accreditation is publicly awarded ($\kappa \geq 0$). All graduates in the early post-treatment window ($\kappa \in \{1, 2, 3, 4\}$) were enrolled before accreditation was announced, so any wage effect in this window cannot be driven by compositional changes in the student body. The composition of the graduating cohort in this early window is determined entirely by enrollment decisions made before accreditation was public.

4.2 Estimation and Inference

The estimation sample is defined in Section 3. The remaining challenge is estimating the effect of accreditation without importing the biases that contaminate the standard two-way fixed effects estimator in staggered settings. When treatment effects are heterogeneous across cohorts, the traditional estimator uses early-treated units as implicit controls for later-treated units, producing estimates that can be negatively weighted and misleading (Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021; Sun and Abraham, 2021). We avoid this entirely by using the imputation estimator proposed by Borusyak et al. (2024).

The imputation approach proceeds in two steps. We first estimate a unit-and-period fixed effects model on the sample of never-treated and not-yet-treated observations:

$$Y_{pt} = \alpha_p + \lambda_t + \varepsilon_{pt} \tag{2}$$

We then use the estimated parameters to impute counterfactual outcomes $\hat{Y}_{pt}^{(0)}$ for treated observations. The average treatment effect on the treated (ATT) for each event-time κ is the mean difference between observed and imputed outcomes:

$$\widehat{\text{ATT}}(\kappa) = \frac{1}{N_\kappa} \sum_{(p,t):t-G_p=\kappa} \left[Y_{pt} - \hat{Y}_{pt}^{(0)} \right] \tag{3}$$

This estimator is valid under arbitrary treatment effect heterogeneity across cohorts and over time. We cluster standard errors at the program level to account for serial correlation in wages within programs.

The event study profile traces the ATT at each relative period, but the headline estimate reported in Table 3 is a scalar ATT that pools across all cohorts and post-treatment event times. Following Borusyak et al. (2024), this scalar is a weighted average of the cohort-by-event-time specific estimates $\widehat{\text{ATT}}(\kappa)$, where the weights are proportional to the number of treated observations contributing to each cohort-by-event-time estimate. Cohorts with more graduates at a given event time receive more weight, so the scalar ATT reflects where most of the identifying variation lies. The scalar ATT summarizes the average wage effect across all post-accreditation cohorts, while the event study profile reveals whether that effect accrues immediately or builds over time.

4.3 Parallel Trends

Parallel trends requires that, absent accreditation, the wage trajectories of eventually-treated and control programs would have evolved identically over time. In this setting, the assumption is plausible because the timing of first accreditation is not driven by pre-existing wage trends. Programs self-select to apply, but the final decision (including whether accreditation is granted and for how long) is determined by an independent CNA commission after a peer review process whose outcome programs cannot fully anticipate.

We assess this empirically by examining the pre-treatment coefficients for $\kappa \in \{-4, -3, -2\}$, normalized to $\kappa = -1$, and test their joint significance with an F-test. Flat pre-trends and failure to reject the null provide the main empirical support for the assumption.

4.4 No Anticipation

The design requires that neither programs nor employers adjusted wage-relevant behavior before the accreditation announcement. Programs may have prepared for the CNA review in other ways, but if those preparations did not translate into changes that employers could observe and price into wages, the pre-announcement wage trajectory remains a valid counterfactual.

On the program side, no anticipation does not require that programs be passive before the accreditation decision. Molina and Valdebenito (2026) documents that programs increase graduation rates in the period preceding the CNA review, pushing near-completers to finish on time because graduation rates are an input to the commission's evaluation. However, this behavior is administrative rather than a quality investment. Programs are not upgrading infrastructure, faculty, or curriculum in ways that employers could independently observe and price into wages. Without visible quality improvements, employers have no basis for adjusting wages before the formal CNA ruling is public.

On the employer side, the concern is subtler. CNA peer visits bring in evaluators from academia and industry who observe the program before the commission rules. In principle, these evaluators could start adjusting wages for the program's graduates before the ruling is public. Two features of the Chilean labor market rule this out. Wages for a program's graduates reflect market-wide beliefs, and no single employer can unilaterally move them. And evaluators participate in a regulatory process, not a recruiting one, giving them no systematic incentive to favor specific programs in their own hiring.

4.5 Limitations of the Research Design

The staggered design identifies the effect of accreditation on average graduate wages at the program level, but the aggregate structure of the data limits how far the results can speak to the underlying channels. Two channels could generate a wage increase after accreditation. Employers may update their beliefs about the program in response to the new public signal, raising wage offers to graduates they would have otherwise discounted. Alternatively, accreditation may attract higher-ability students whose wages would be higher regardless of where they attended, raising the program's average wages through a change in the composition of the graduating cohort. The early post-accreditation window ($\kappa \in \{1, 2, 3, 4\}$) provides partial separation, as graduates in this window enrolled before accreditation was announced and could not have been drawn by the signal. Figure 1 illustrates this timing structure for a representative five-year program. Any wage effect in the early window cannot be driven by compositional changes in the student body. In the late window ($\kappa \geq D$, where D is program duration), both channels operate simultaneously, and the aggregate data cannot distinguish between them. Separating employer updating from mechanical composition at longer horizons would require individual-level wage records that allow controlling for each graduate's own ability, which falls outside the scope of the current analysis.

Three additional limitations deserve acknowledgment. First, the estimation sample aggregates outcomes by graduation cohort rather than by enrollment cohort. Students who entered a program before accreditation but took longer than expected to graduate appear in the same graduation cohort as students who entered after accreditation and graduated on time. These two groups experienced different selection environments, and grouping them together attenuates the separation between the early and late windows. Disaggregating by enrollment cohort is feasible with the available education data but has not been implemented in the current analysis.

Second, if accreditation draws high-ability students into treated programs, those students are simultaneously leaving untreated programs. The control group may therefore experience a deterioration in graduate quality at the same time that treated programs improve, inflating the estimated treatment effect by moving both sides of the comparison. Appendix Table A1 restricts the control group to not-yet-treated programs only, excluding the 773 never-treated programs that may be most susceptible to student outflows. The results are qualitatively similar and larger in magnitude, suggesting that the baseline estimates are not inflated by general equilibrium contamination of the never-treated control group.

Third, the wage data cover private-sector payroll exclusively, as discussed in Section 3. If accreditation shifts graduates across sectors, the private-sector sample captures only part of the labor market response, and the direction of the bias depends on whether the graduates who move into the public sector are positively or negatively selected on ability.

5 Results

5.1 Graduation Counts and Employment Rate

Any effect of accreditation on average graduate wages could reflect either a change in how the labor market values graduates from the program or a change in who graduates from the program. If accreditation alters the size or composition of graduating cohorts, the average wage of graduates shifts even if no individual graduate's wage changes. Establishing whether and how the graduate pool changes is therefore a prerequisite for interpreting the wage estimates. We estimate event studies for the log of total graduates per program–graduation year and the log of graduates found in private-sector formal employment records, along with the employment rate (the ratio of the two). The first two outcomes are necessary to correctly interpret the third, as changes in the employment rate can be driven by the numerator, the denominator, or both. Figure 3 presents all three event studies.

Treated programs produce fewer graduates than control programs in the years before accreditation. The pre-period pattern violates parallel trends for the graduation count outcome, and the violation points directly to a known mechanism. Programs preparing for a peer review face strong incentives to demonstrate completion rates, which encourages delayed completers to finish before the self-evaluation report is submitted or the peer evaluation visit conducted. Molina and Valdebenito (2026) documents this gaming behavior for the full population of accrediting programs. The anticipation effect operates before the formal decision, not after it. After accreditation is granted, total graduate counts rise. Students who had completed coursework but delayed their final requirement, typically a thesis or capstone project, may respond to accreditation by completing their degree if they believe the newly certified status improves their labor market prospects. Programs themselves may lower completion barriers for near-completers once the accreditation evaluation is behind them. The aggregate data cannot distinguish between these explanations, but both are consistent with accreditation expanding the pool of graduates beyond what the pre-accreditation trend would predict.

The count of graduates found in private-sector formal employment records also rises after accreditation, but less markedly than total graduate counts. This asymmetry is the key to interpreting the employment rate. Both series increase, but the pool of completers expands faster than the pool of formal-sector workers, which is sufficient to pull the employment rate down even when employment counts are rising. Before accreditation, treated programs exhibit higher formal employment rates than control programs. After accreditation, formal employment rates fall toward or below control levels. This is not evidence that accreditation hurts labor market outcomes. The additional completers who graduate after receiving the certification could also be more likely to enter self-employment, the informal sector, or public-sector positions that fall outside the SII payroll records, so the formal-employment count rises at a slower pace than total graduates.

5.2 Enrollment Timing and the Composition of Graduates

The graduation count results establish that the size of the graduate pool changes after accreditation. The next question is whether the composition of the pool changes as well. If accreditation attracts higher-ability students, the composition of graduating cohorts should shift upward, but only after those students have had time to complete a multi-year degree.

The timing argument in Section 4 provides a first, indirect answer. Students who enroll because a program is accredited enter a multi-year degree and graduate no earlier than $\kappa = D$, where D is program duration. At $\kappa \in \{1, 2, 3, 4\}$, the composition of the graduate pool is entirely determined by enrollment decisions made before accreditation was public.[2] The composition channel cannot operate in this early window. At $\kappa \geq D$, both employer updating and compositional changes operate together, and any wage effect at those horizons reflects both.

Table 3 and Figure 4 provide a direct test of this compositional shift. The ATT for the log-ratio of high-ability graduates (graduates scoring in the top quartile of the PSU) rises by 0.121 log points after accreditation ($p < 0.01$). Treated programs attract more high-scoring students once the accreditation stamp is on the record. The timing of this shift, shown in Figure 4, is informative. Students who enrolled because of the accreditation signal only enter the labor market at $\kappa \geq D$, so the ability shift builds gradually and accounts for a growing share of the wage effect at longer horizons. High-ability students prefer certified programs because the credential amplifies their own labor market prospects, and the resulting enrollment shift appears only after accreditation is public and only in graduating cohorts several years later.

The income composition measure shows no significant change after receiving the certification (-0.047, $p > 0.10$). Accreditation shifts the ability composition of graduates but not their socioeconomic background.

5.3 Average Effect of Accreditation on Employed Graduate Wages

With the composition and graduation count patterns established, we turn to the wage estimates. Table 3 reports the scalar average treatment effect on the treated (ATT) at each wage horizon. The estimated effect is 3.0 percent for wages measured one year after graduation, 2.9 percent for wages measured two years after graduation, and 2.4 percent for wages measured three years after graduation. The first-year estimate is imprecise and not statistically distinguishable from zero. The year-two and year-three estimates are statistically significant at the 10 percent level. In absolute terms, the year-two effect corresponds to roughly CLP 234,000 in additional annual earnings against a control mean of

[2]Figure 7 shows the full distribution of program duration in Chile. The average duration of a degree program is 9.4 semesters, with more than 70 percent of programs lasting between 8 and 11 semesters (four to five years). This confirms that $\kappa = 5$ is the first post-accreditation period at which students who enrolled after the accreditation announcement begin graduating.

CLP 8.1 million, and the year-three effect corresponds to roughly CLP 235,000 against a control mean of CLP 9.8 million.

The event study in Figure 2 shows the full dynamic profile. Pre-period coefficients at $\kappa \in \{-4, -3, -2\}$ are statistically indistinguishable from zero and jointly insignificant ($p > 0.60$), supporting the parallel trends assumption. For cohorts graduating after accreditation, the wage profile diverges and stays above the pre-trend baseline. In the early post-accreditation window, individual effects at $\kappa \in \{1, 2, 3, 4\}$ are close to zero. At longer horizons ($\kappa \in \{5, 6, 7, 8\}$), the estimates reach 6 to 15 percent.

The timing of the wage effects is informative when read alongside the composition evidence in Section 5.2. In the early window, graduates enrolled before accreditation was awarded and the composition of the graduating cohort has not yet changed, so the near-zero estimates in this window are consistent with limited immediate wage response. In the late window, higher-ability cohorts who self-selected into the now-accredited program begin graduating, and the wage estimates rise accordingly. The aggregate data cannot determine how much of the late-window effect reflects employer responses to the accreditation signal versus the mechanical consequence of higher-ability graduates earning more. The wage estimates are constructed from the private-sector formal employment sample, so graduates who enter the public sector, self-employment, or the informal sector are not observed.

The 3 percent pooled ATT is therefore a weighted average: a near-zero effect in the early window, where the composition of graduates has not yet shifted, and a growing effect at longer horizons, where higher-ability cohorts account for an increasing share of program graduates. The heterogeneity analysis examines whether this effect varies with institutional reputation and program accreditation intensity.

6 Heterogeneity

6.1 Employer Prior Beliefs and Institutional Reputation

The average wage effect documented in Section 5 pools across programs where the accreditation signal may carry very different informational content. Institutional reputation provides a natural source of variation in how much employers already knew about a program before the accreditation decision. Employers have accumulated information about programs at well-established universities through decades of direct hiring and word-of-mouth. Programs at less prominent institutions are harder to evaluate from reputation alone.

Table 4 reports subgroup ATTs stratified by the institution's accreditation tier. Graduates of programs at *Baseline* institutions realize a wage increase of 9.1 percent in their first year of employment ($p < 0.01$), falling to 4.8 percent in year two and 3.0 percent in year three. At the control mean of CLP 6.1 million for first-year wages, the 9.1 percent effect amounts to roughly CLP 550,000 per year.

For graduates of programs hosted at *Enhanced* institutions, the first-year effect is -2.6 percent ($p < 0.05$). For graduates of *Top-tier* institutions, the first-year effect is -4.8 percent ($p < 0.01$), remaining negative through year two (-2.6 percent) and year three (-2.0 percent). At the control mean of CLP 7.1 million for first-year wages among top-tier graduates, the first-year reduction corresponds to roughly CLP 340,000 per year. The wage response does not merely fade as institutional prestige rises; it reverses sign. Figure 5 traces the diverging event study profiles across tiers.

6.2 Content of the Accreditation Signal

Institutional reputation captures what employers already believed about the program's host institution. The accreditation term awarded to the program conveys additional program-specific information. A short initial term certifies quality at the minimum standard. A long initial term certifies quality well above it. As documented in Section 2 and Table 2, these two dimensions do not perfectly correlate.

Table 5 reports subgroup ATTs stratified by the duration of the first accreditation term. Programs receiving a *Low*-intensity first accreditation (1–3 years) show a first-year wage effect of 12.3 percent ($p < 0.01$). At the control mean of CLP 6.2 million for first-year wages, this amounts to roughly CLP 760,000 in additional annual earnings. The effect remains statistically significant through the third year: 8.2 percent in year two and 6.4 percent in year three.

Programs receiving a *Mid*-intensity first accreditation (4–5 years) show a first-year wage effect of -4.5 percent ($p < 0.01$). Programs receiving a *High*-intensity term (6–7 years) show no statistically significant change (-1.2 percent, $p > 0.10$). Figure 6 presents the event study profiles for all three intensity groups.

The two sets of results point in the same direction. The wage response is largest for programs at institutions where employers had the least prior information and for programs that received the shortest accreditation terms. The wage response is negative or zero for programs at institutions where employer beliefs were already well-established and for programs that received longer terms.

6.3 Limitations of the Heterogeneity Analysis

The heterogeneity results presented above examine each dimension separately. However, as Table 2 documents, the institutional tier and program accreditation term are not perfectly aligned. Nearly 30 percent of programs at Top-tier institutions received a Low first-accreditation term, and programs at Baseline institutions span the full range of accreditation outcomes. The wage response in any given subgroup therefore reflects a mixture of prior beliefs and signal content that the separate analyses cannot disentangle. A fully interacted specification that estimates the wage effect for each combination of institutional tier and program accreditation term would isolate where the effects concentrate, and preliminary results suggest that specific combinations drive the aggregate patterns. This analysis requires careful consideration of the appropriate grouping structure and is left for future work.

More broadly, the heterogeneity analysis inherits the limitations of the aggregate research design discussed in Section 4. Within each subgroup, the aggregate data cannot separate employer responses to the accreditation signal from the mechanical effect of graduating higher-ability students. Individual-level wage records that allow controlling for each graduate's own ability would permit this decomposition within each institutional tier and program intensity group, providing a sharper test of how much of the heterogeneity reflects differential employer updating versus differential composition shifts across subgroups. Similarly, the aggregate data report only mean wages per program-cohort, so the analysis cannot examine whether accreditation compresses or widens the within-cohort wage distribution differently across subgroups. These extensions require individual-level data and are left for future work.

7 Conclusion

We study what happens to graduate wages when a university program receives its first accreditation from Chile's National Accreditation Commission. The analysis uses a staggered difference-in-differences design, estimated with the imputation approach of Borusyak et al. (2024), and documents three sets of findings.

First, the size and composition of the graduate pool change after accreditation. Total graduation counts rise, while the formal employment rate falls as the expanded pool includes graduates who do not appear in private-sector payroll records. Supplementary event studies show that programs inflate graduation counts in anticipation of the peer evaluation, and the pre-period pattern for this outcome violates parallel trends, consistent with strategic behavior around the accreditation process. The ability composition of graduating cohorts shifts upward, with the share of high-ability graduates (top quartile of entrance exam scores) increasing significantly after accreditation ($p < 0.01$). The socioeconomic composition does not change. These patterns are consistent with accreditation attracting higher-ability students into treated programs, with the compositional shift appearing in graduating cohorts several years after the accreditation decision, once those students have completed their degrees.

Second, the average wage effect across all post-accreditation cohorts is approximately 3 percent, with statistical significance in years two and three after graduation. Pre-trend tests confirm that wages at treated and control programs followed parallel trajectories before accreditation ($p > 0.60$ across all three wage horizons), supporting the identification assumptions for the wage outcomes. The dynamic profile reveals that this average masks substantial variation over time. In the early post-accreditation window ($\kappa \in \{1, 2, 3, 4\}$), when all graduates enrolled before accreditation was announced and the composition of the graduating cohort has not yet changed, the wage estimates are close to zero. At longer horizons ($\kappa \geq 5$), where higher-ability cohorts begin graduating, the estimates rise to 6 to 15 percent. The aggregate data cannot determine how much of the late-window effect reflects employer responses to the accreditation signal versus the mechanical consequence of higher-ability graduates earning more.

Third, the wage effect varies substantially across two dimensions. When stratified by institutional reputation, graduates of programs at *Baseline* institutions experience wage increases of 9.1 percent in year one, while graduates of programs at *Top-tier* institutions experience wage reductions of 4.8 percent. When stratified by program accreditation intensity, programs receiving a *Low* first-accreditation term (1–3 years) show wage increases of 12.3 percent in year one, while programs receiving a *High* term (6–7 years) show no significant change. Both dimensions point in the same direction: the wage response is largest where prior information about the program was scarce and the accreditation term was short, and it is negative or absent where prior information was abundant.

Several limitations of the current analysis should inform how these results are interpreted. The wage data cover private-sector formal employment only, so graduates who enter the public sector, self-employment, or the informal sector are not observed. The analysis aggregates outcomes by graduation cohort rather than enrollment cohort, which attenuates the separation between the early and late post-accreditation windows. The two dimensions of heterogeneity are examined separately, but the cross-tabulation of institutional tier and program accreditation term (Table 2) shows that they do not perfectly align, and the wage response may concentrate in specific combinations that the separate analyses cannot isolate. Individual-level wage records that allow controlling for each graduate's own ability would permit a sharper decomposition of employer responses from mechanical composition effects within each subgroup, and would allow examining whether accreditation compresses or shifts the within-cohort wage distribution. These extensions are left for future work.

Table 1: Descriptive Statistics: Pre-Accreditation Cohorts

	Treated	Control	Difference
Panel A: Labor Market Outcomes			
Graduation cohort size	32.735	24.543	8.192***
	(38.245)	(40.087)	(0.865)
Employment rate, year 1	0.821	0.793	0.028***
	(0.233)	(0.246)	(0.005)
Annual earnings, yr. 1 (CLP)	5969.989	7610.285	-1640.296***
	(3595.342)	(4565.623)	(91.375)
Panel B: Student Characteristics			
PSU score (math-verbal)	521.910	519.340	2.570
	(90.097)	(73.376)	(1.897)
GPA score	582.840	562.614	20.227***
	(66.823)	(61.125)	(1.482)
Male	0.412	0.547	-0.135***
	(0.320)	(0.312)	(0.007)
High income	0.169	0.161	0.009**
	(0.209)	(0.196)	(0.004)
College-educated parents	0.456	0.403	0.053***
	(0.244)	(0.233)	(0.005)
Panel C: Institutional Quality			
Baseline Inst.	0.365	0.374	-0.009
	(0.481)	(0.484)	(0.011)
Enhanced Inst.	0.529	0.454	0.076***
	(0.499)	(0.498)	(0.011)
Top-tier Inst.	0.106	0.172	-0.066***
	(0.308)	(0.378)	(0.008)
N (Programs)	1154	773	

Note: This table displays the means of program-level characteristics for the Control (programs never accredited during the sample period) and Treated (eventually-accredited programs, observed in pre-accreditation cohorts) groups. Standard deviations are reported in parentheses beneath the means. The unit of observation is the academic program. The "Difference" column reports the coefficient from an OLS regression of the variable on a treatment status indicator. The N (Programs) row counts programs with at least one pre-accreditation wage observation. Institution quality tiers are defined by CNA accreditation duration: Baseline (1–3 years), Enhanced (4–5 years), Top-tier (6–7 years). Significance levels are denoted as follows: * p<0.1, ** p<0.05, *** p<0.01.

Table 2: Distribution of Treated Programs by Institutional Tier and Accreditation Term

	Program First-Accreditation Term			
	Low	Mid	High	Total
Baseline	426 (58.3%)	242 (33.1%)	63 (8.6%)	731
Enhanced	301 (40.1%)	408 (54.3%)	42 (5.6%)	751
Top-tier	52 (29.7%)	88 (50.3%)	35 (20.0%)	175

Note: This table displays the joint distribution of treated programs by institutional accreditation tier (rows) and program first-accreditation term (columns). Each entry reports the count and row percentage of programs in each combination. Institutional tier is the modal CNA accreditation duration of the host institution: Baseline (1–3 years), Enhanced (4–5 years), Top-tier (6–7 years). Program term is the duration awarded at first program-level accreditation: Low (1–3 years), Mid (4–5 years), High (6–7 years).

Table 3: Effect of First-Time Accreditation on Graduate Wages and Graduate Composition

	Labor Market Outcomes (logs)			Grads composition	
	Wage yr 1	Wage yr 2	Wage yr 3	Ability	Income
Accreditation effect	0.030	0.029*	0.024*	0.121***	-0.047
	(0.022)	(0.017)	(0.014)	(0.046)	(0.042)
Num. Obs	18672	18707	18708	13443	13217
Control mean	6012.6	8078.4	9806.1	0.4	0.2

Note: Standard errors (clustered at the program level) are in parentheses. The control outcome mean for wage columns is the average annual wage among never-accredited programs, reported in thousands of CLP. Composition outcomes are log-ratios: $\log(s/(1-s))$ where s is the program-cohort share of high-PSU graduates (top quartile, year-specific cutoff) or high-income graduates. Composition outcomes are available only for graduates matched to DEMRE centralized admissions records. All programs are offered by accredited institutions. The treatment group includes programs accredited for the first time. Average treatment effects on the treated follow Borusyak et al. (2024). * p<0.1, ** p<0.05, *** p<0.01.

Table 4: Heterogeneous Effects by Institutional Accreditation Quality

	Log wage yr 1	Log wage yr 2	Log wage yr 3
Accreditation × Baseline inst.	0.091***	0.048***	0.030***
	(0.017)	(0.011)	(0.009)
Accreditation × Enhanced inst.	-0.026**	-0.001	0.013*
	(0.010)	(0.009)	(0.008)
Accreditation × Top-tier inst.	-0.048***	-0.026**	-0.020**
	(0.011)	(0.010)	(0.008)
Num. Obs	17978	18015	18013
Control Mean (CLP thousand)			
Baseline inst.	6057.3	7936.2	9592.7
Enhanced inst.	5996.4	8124.3	9966.6
Top-tier inst.	7135.4	9316.7	11080.8

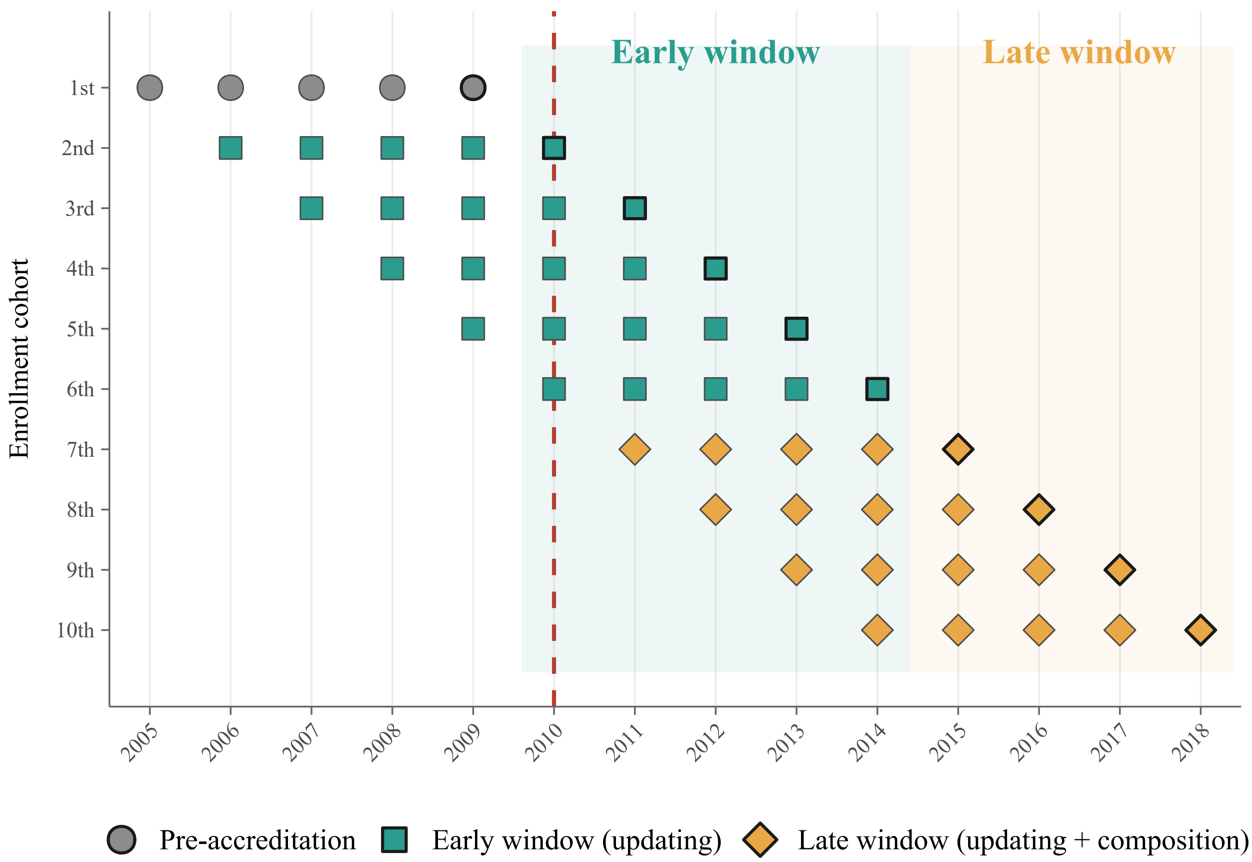
Note: This table displays heterogeneous effects of first-time accreditation on graduate wages by institutional quality. Standard errors (clustered at the program level) are in parentheses. The unit of observation is the academic program. Each row reports the average treatment effect weighted by the share of treated observations within that institutional tier, estimated following Borusyak et al. (2024) with program and graduation-year fixed effects. Institutional quality is defined by the modal CNA accreditation duration of the host institution: Baseline (1–3 years), Enhanced (4–5 years), Top-tier (6–7 years). Control means are in CLP thousands and correspond to never-accredited programs within each institutional tier. Significance levels are denoted as follows: * p<0.1, ** p<0.05, *** p<0.01.

Table 5: Heterogeneous Effects by Program Accreditation Intensity

	Log wage yr 1	Log wage yr 2	Log wage yr 3
Accreditation × Low (1-3 years)	0.123***	0.082***	0.064***
	(0.027)	(0.019)	(0.015)
Accreditation × Mid (4-5 years)	-0.042***	-0.015	-0.004
	(0.012)	(0.011)	(0.009)
Accreditation × High (6-7 years)	-0.012	-0.022*	-0.006
	(0.014)	(0.012)	(0.012)
Num. Obs	17978	18015	18013
Control mean (CLP thousand)	6202.9	8245.9	10006.5

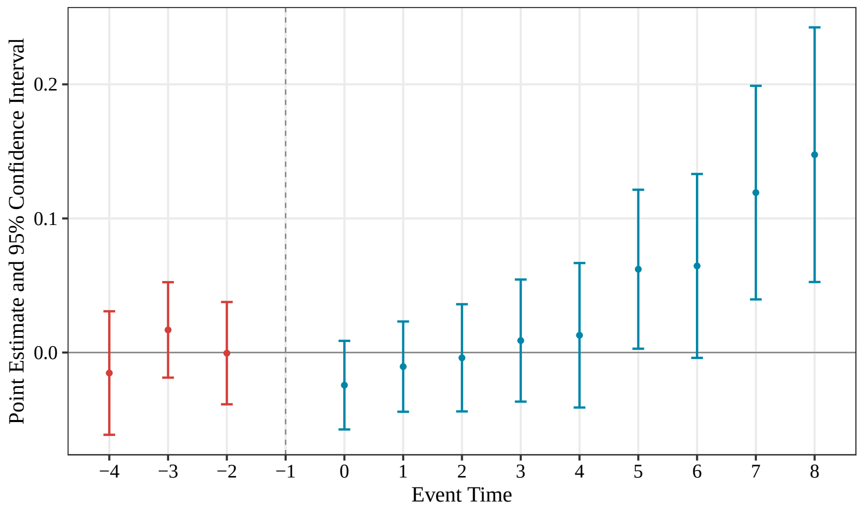
Note: This table displays heterogeneous effects of first-time accreditation on graduate wages by program accreditation intensity. Standard errors (clustered at the program level) are in parentheses. The unit of observation is the academic program. Each row reports the average treatment effect weighted by the share of treated observations within that program intensity tier, estimated following Borusyak et al. (2024) with program and graduation-year fixed effects. Program intensity is defined by the duration of the first accreditation term awarded to the program: Low (1–3 years), Mid (4–5 years), High (6–7 years). The control mean corresponds to the full pool of never-accredited programs, as program accreditation intensity is a treatment outcome and cannot be observed for control programs. Significance levels are denoted as follows: * p<0.1, ** p<0.05, *** p<0.01.

Figure 1: Enrollment cohorts and the accreditation event

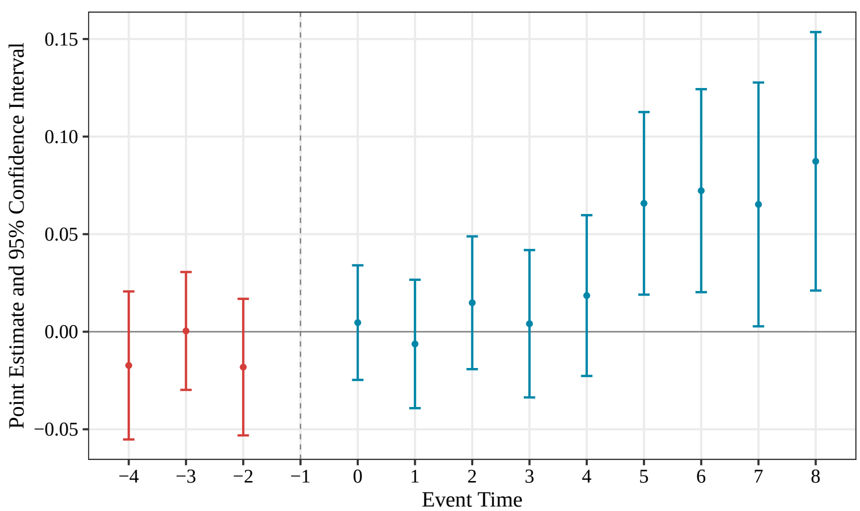


Notes. Timing structure for a representative program with a five-year duration that receives its first accreditation in 2010. Each marker represents a cohort-year. Circles denote pre-accreditation cohorts that enrolled and graduated before the accreditation decision. Squares denote early-window cohorts that enrolled before accreditation but graduated after the decision was public. Diamonds denote late-window cohorts that enrolled after accreditation was announced. Larger markers with heavier borders indicate expected graduation years. In the early window, the composition of the graduating cohort has not yet changed. This becomes true in the late window.

Figure 2: Event study estimates: Wages of employed grads



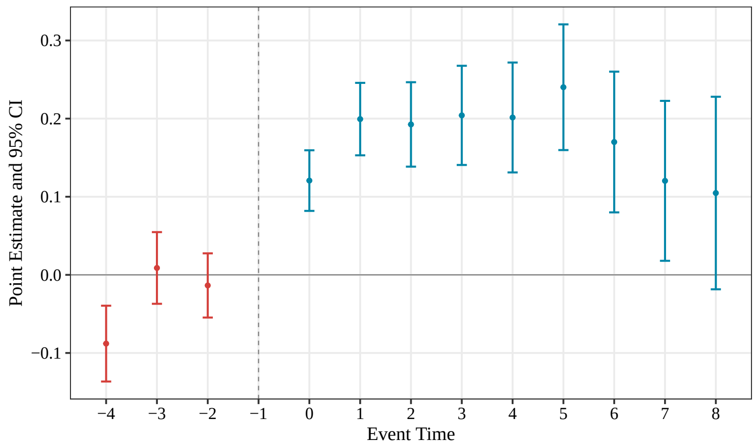
(a) $1^{st} Year$



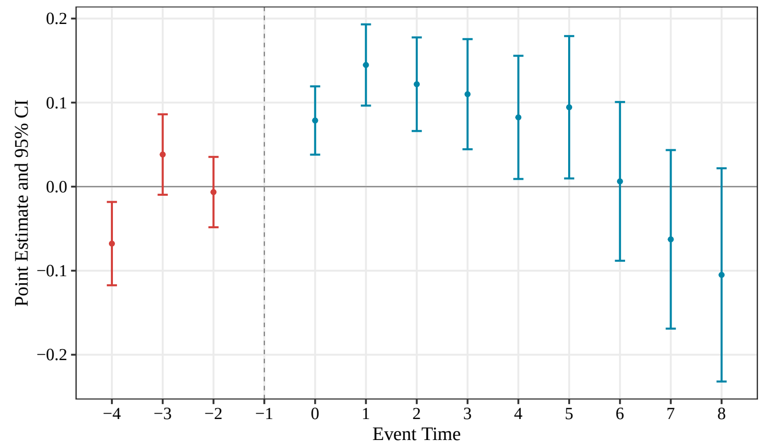
(b) $2^{nd} Year$

Note: This figure plots event study estimates of the effect of first-time accreditation on log wages of graduates (conditional on being employed): (a) 1 year, and (b) 2 years after graduation. $\kappa = -1$ is the reference period. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

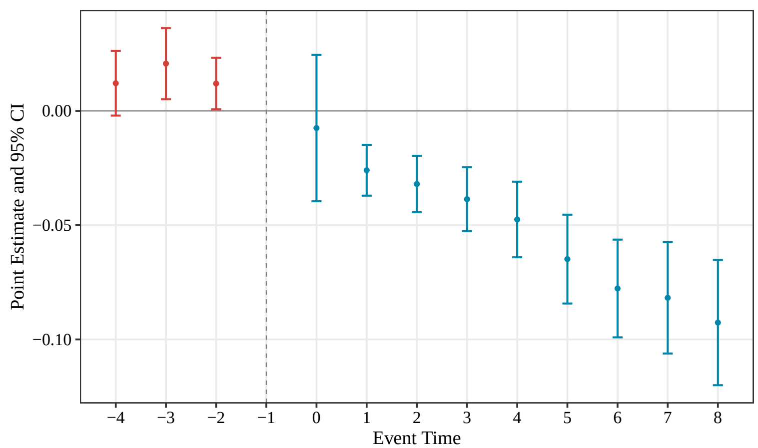
Figure 3: Event study estimates: Mechanism outcomes



(a) Graduation cohort size



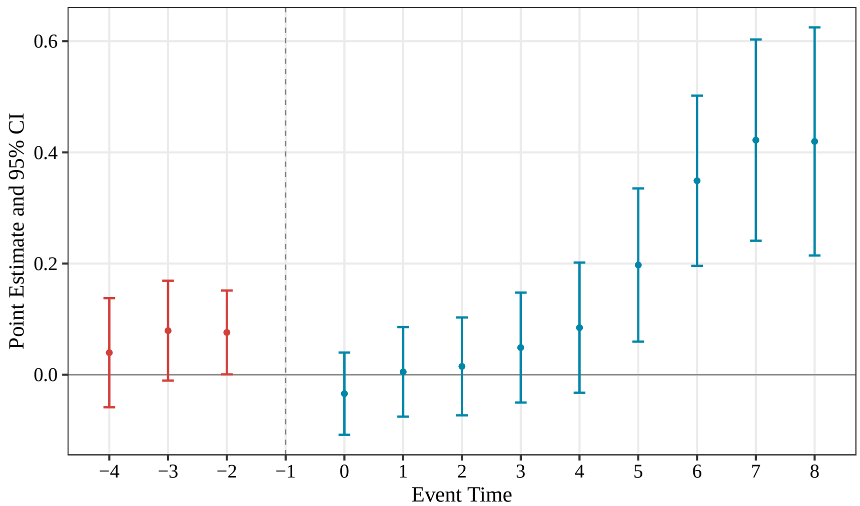
(b) Employed cohort size



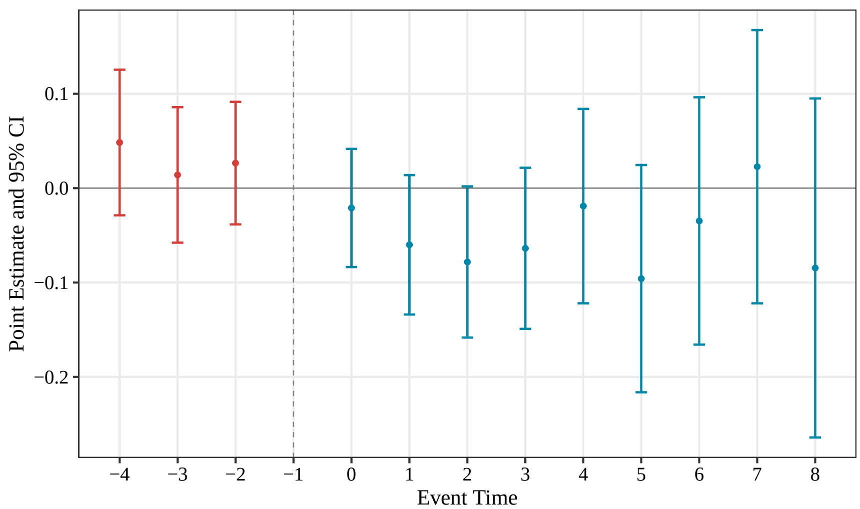
(c) Employment rate

Note: This figure plots event study estimates of the effect of first-time accreditation on three mechanism outcomes: (a) log graduation cohort size (graduates per program–graduation year), (b) log employed cohort size (graduates matched to private-sector earnings records), and (c) formal employment rate two years after graduation. These are descriptive estimates as pre-trend tests reject parallel trends for all three outcomes, consistent with anticipatory increases in on-time graduation rates documented in Molina and Valdebenito (2026). $\kappa = -1$ is the reference period. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

Figure 4: Event study estimates: Graduate composition following accreditation



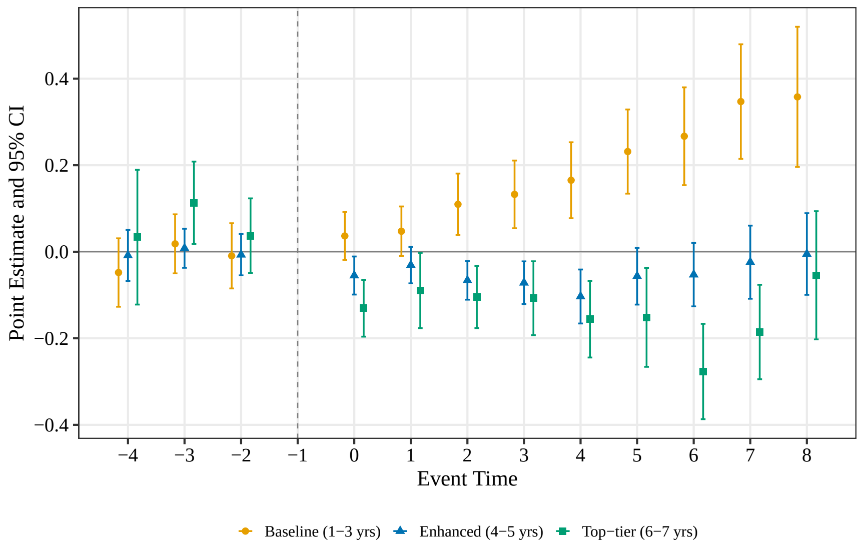
(a) Academic ability composition



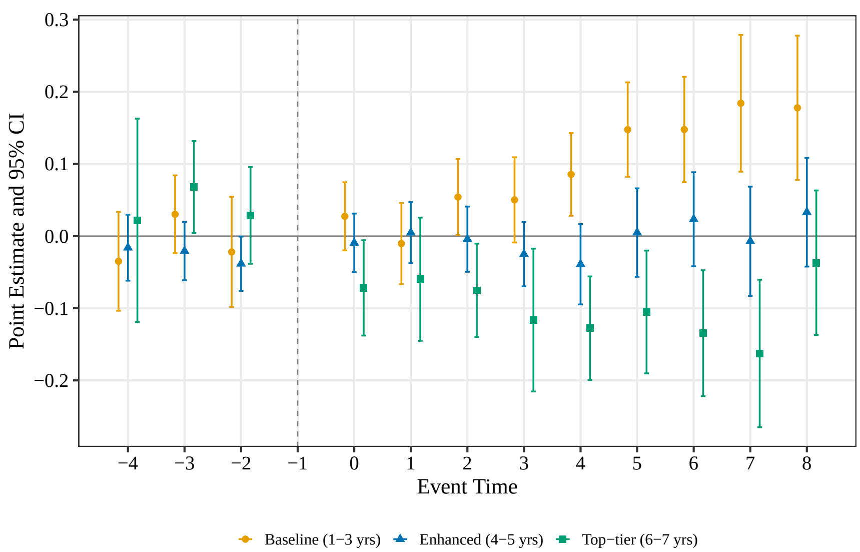
(b) Socioeconomic composition

Note: This figure plots event study estimates of the effect of first-time accreditation on the log-ratio of high- to low-group graduates within each program–graduation year. Panel (a) defines high ability as scoring at or above the year-specific 75th percentile of the distribution of average DEMRE verbal and math scores. Panel (b) defines high socioeconomic status as belonging to a household with an above-median household income. A positive estimate indicates that the graduating cohort tilts toward higher-ability or higher-income graduates relative to the counterfactual. $\kappa = -1$ is the reference period. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

Figure 5: Event study estimates: Wages of employed grads by institutional tier



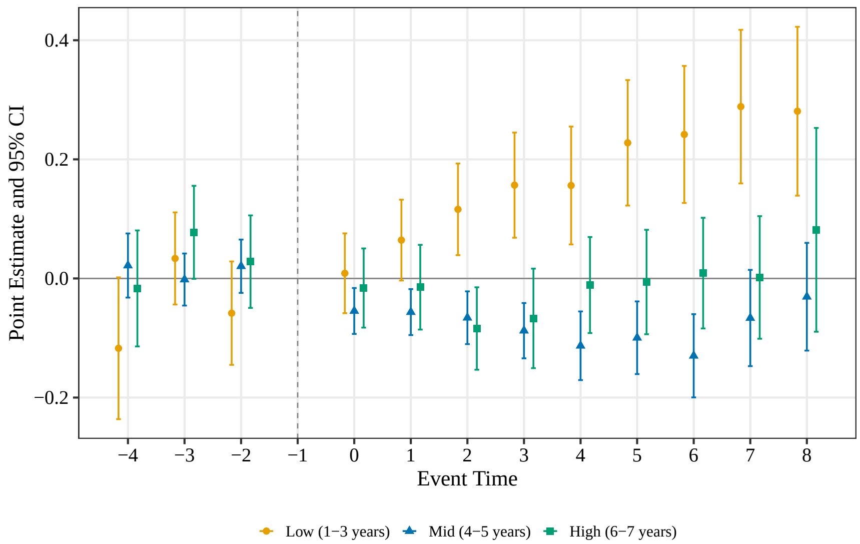
(a) $1^{st} Year$



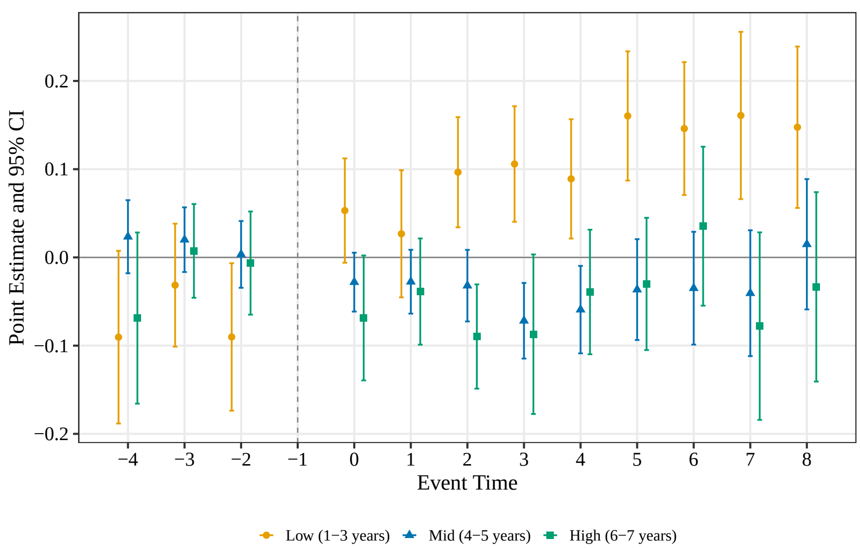
(b) $2^{nd} Year$

Note: This figure plots event study estimates of the effect of first-time accreditation on log wages of graduates by institutional quality tier. Panels (a) and (b) show estimates for wages after 1 and 2 years after graduation. Colors and marker shape indicate institutional quality tiers: baseline (yellow), enhanced (blue), top tier (green). $\kappa = -1$ is the reference period. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

Figure 6: Event study estimates: Wages of employed grads, by treatment intensity



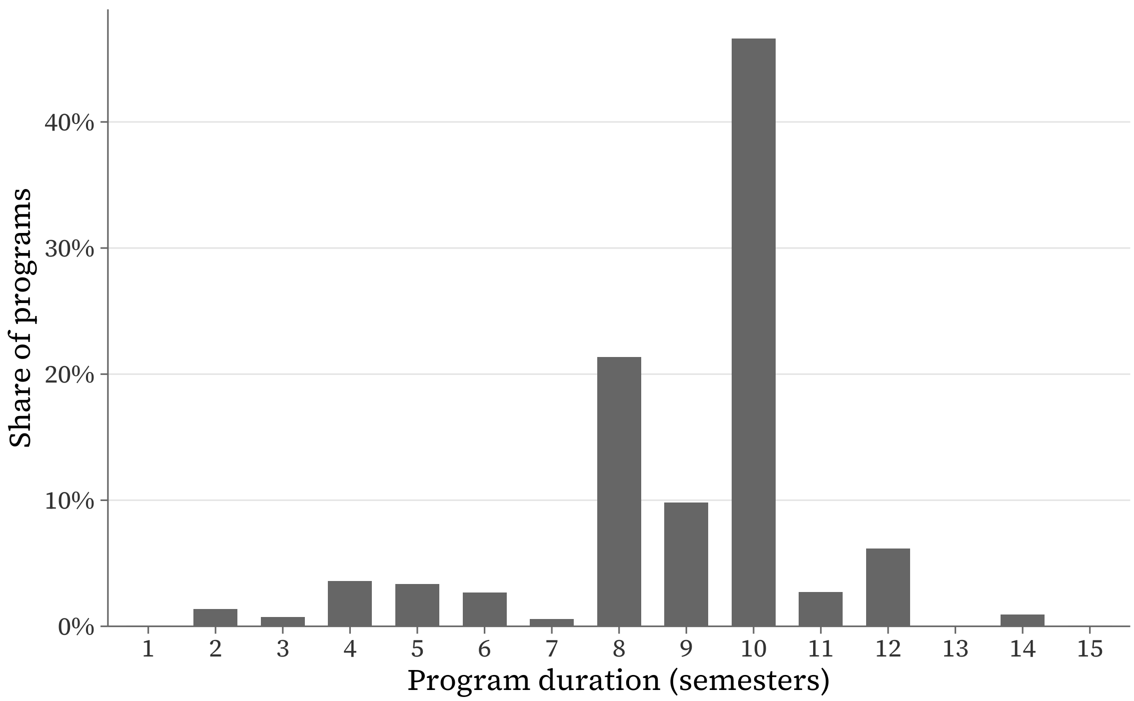
(a) $1^{st} Year$



(b) $2^{nd} Year$

Note: This figure plots event study estimates of the effect of first-time accreditation on log wages of graduates by program treatment intensity. Panels (a) and (b) show estimates for wages after 1 and 2 years after graduation. Colors and marker shape indicate first-accreditation intensity tier: low (yellow), mid (blue), high (green). $\kappa = -1$ is the reference period. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

Figure 7: Distribution of programs duration



Note: This figure plots the full distribution of programs' duration in the Chilean higher education system. more than 70% of programs have a duration of between 4 and 5 years.

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A Robustness Checks

A.1 Not-Yet-Treated Control Group

The baseline specification uses both never-treated and not-yet-treated programs as controls for the imputation estimator. If accreditation draws high-ability students into treated programs, those students leave the programs they would have otherwise attended. Never-treated programs, which by definition never receive accreditation, may be the most exposed to this outflow because they cannot offset it by attracting signal-responsive students of their own. The resulting deterioration in the never-treated control group's graduate composition would inflate the estimated treatment effect by moving both sides of the comparison.

To assess the sensitivity of the results to this concern, we re-estimate the main specification restricting the control group to not-yet-treated programs only. These programs will eventually receive accreditation themselves, so they may be less susceptible to permanent student outflows than never-treated programs. Table A1 reports the pooled ATT under both control group definitions. The not-yet-treated estimates are qualitatively similar to the baseline and, if anything, larger in magnitude. The year-two estimate rises from 2.9 percent ($p < 0.10$) to 7.4 percent ($p < 0.01$) when never-treated programs are excluded. This pattern is inconsistent with general equilibrium contamination, inflating the baseline estimates. If anything, the inclusion of never-treated programs in the control group produces more conservative results, suggesting that never-treated programs serve as a stable comparison group for the wage outcomes examined here.

Table A1: Robustness: Not-Yet-Treated Programs as Control Group

	Log wage yr 1	Log wage yr 2	Log wage yr 3
Panel A: Full control group (never-treated + not-yet-treated)			
Accreditation effect	0.030	0.029*	0.024*
	(0.022)	(0.017)	(0.014)
Num. Obs	18672	18707	18708
Panel B: Not-yet-treated controls only			
Accreditation effect	0.053	0.074***	0.040
	(0.033)	(0.027)	(0.025)
Num. Obs	14258	14293	14284

Note: This table compares the pooled average treatment effect of first-time accreditation on graduate wages across two control group definitions. Standard errors (clustered at the program level) are in parentheses. Panel A uses the full control group (never-treated and not-yet-treated programs). Panel B restricts the control group to not-yet-treated programs only, excluding the 847 never-treated programs to address potential general equilibrium effects on the composition of never-treated controls. Estimated following Borusyak et al. (2024) with program and graduation-year fixed effects. Significance levels are denoted as follows: * p<0.1, ** p<0.05, *** p<0.01.